

Chapter 123 — Sphincteric Incontinence in the Male: AUS and Perineal Slings — Numbers, Gold Standards and Traps

IF YOU READ NOTHING ELSE

- **Artificial urinary sphincter (AUS) = gold standard**, all degrees of sphincteric urinary incontinence (SUI)
- **Severe UI or any radiotherapy (RT) → AUS first-line** (AUA/SUFU)
- **Sling = mild-moderate**: 24-h pad weight mild <150 g, moderate <400 g
- **Pressure-regulating balloon (PRB)**: 61–70 bulbar; 51–60 high-risk; 71–80 bladder neck (cm H₂O)
- **Bulbar cuff proximal to corporal fusion**; usually 4.0–4.5 cm
- **Deactivate 4–6 weeks** after implant
- **Erosion risk**: RT, prior erosion, urethroplasty (urethroplasty ≥ RT)
- **Infection or erosion → explant everything**; reimplant 3–6 months
- **Device >3 years old failing → replace whole AUS** (PRB)
- **Failed second void trial at 48 h → suprapubic catheter**, not urethral

AETIOLOGY

- **Male SUI needs both sphincters impaired**; **commonest cause**: radical prostatectomy (RP)
- **Prevalence**: 11% at 60–64; 31% >85
- **Continence improves up to 2 years** post-RP
- **Nerve sparing → faster, not better final**, continence
- **Climacturia 20–68%** post-RP (includes arousal leakage)

MECHANISM OF DEVICES

- **AUS**: circumferential compression, cuff ~2 cm long; voids by straining or **Credé**
- **Advance XP (transobturator)**: repositions and lengthens membranous urethra — **no compression**; needs coapting sphincter
- **Virtue**: quadratic — prepubic **compression** + transobturator elevation
- **ATOMS**: cushion compression; **scrotal port** adjustment
- **ProACT**: two **bladder-neck balloons**, lateral compression; scrotal ports

EVALUATION

- **Core**: history, examination, **postvoid residual (PVR)**, urinalysis
- **Most important history point**: **classify UI** (stress, urge, mixed, overflow)
- **Voiding diary 2–7 days**; **pad test 24 h**
- **Cough test grades**: 0 reported only · 1 delayed drops · 2 early drops · 3 delayed stream · 4 early persistent stream
- **Cystoscopy mandatory** if stricture, vesicourethral anastomotic stenosis (VUAS) or bladder neck contracture suspected
- **Poorly coapting sphincter on cystoscopy → AUS**, not sling
- **Urodynamics (UDS) not routine**: mixed UI, SUI not shown, overflow, poor compliance
- **Detrusor hypocontractility → AUS**
- **Urinalysis ± culture mandatory**; eradicate UTI pre-op

INDICATIONS FOR SURGERY

- **Operate if**: irreversible sphincter incompetence + UI affecting quality of life
- **Severe or gravitational SUI unimproved at 6 months → operate**; don't wait
- **Pelvic floor muscle training**: **no benefit beyond 12 months**
- **First-line**: **AUS and perineal sling**; bulking agents only if small-volume and unfit
- **Absolute (need non-SUI alternative)**: low-compliance bladder threatening kidneys; low-capacity bladder; tissue unfit for any implant

- **Relative**: frequent future transurethral work — bladder cancer, refractory VUAS, radiation cystitis
- **Radiation cystitis**: hyperbaric O₂, **6 months bleed-free** before AUS
- **No AUS**: poor dexterity, cognitive impairment, poor urethra, severely low compliance
- **AUS**: cure ~56%; social continence (0–1 pad) 55–77%; satisfaction ~90%; revision 10–29% at 3 years
- **Transobturator sling cure ~60%**; **ATOMS ~67% dry**; **ProACT ~60% dry**
- **MASTER RCT**: sling non-inferior to AUS overall; **AUS favoured >250 g or post-RT**
- **Sling failure predictors**: RT, severity, previous surgery
- **AUS lifespan 8–10 years** (AUA/SUFU); **expect one thin pad/day**

HIGH-RISK PATIENT

- **Erosion**: RT 3–33% vs non-RT 1–13%; RT erodes sooner (~1 vs ~3 years)
- **Prior erosion → 5-year re-erosion ~50%**
- **Age >80, coronary artery disease → erosion**; diabetes and smoking not shown independently
- **Low testosterone predicts erosion**; replacement unproven
- **Transcorporal (TC) AUS**: **Webster 2002** — tunica layer over dorsal urethra

TECHNIQUE

- **Antibiotics within 60 min**: aminoglycoside 5 mg/kg + cephalosporin or vancomycin
- **Cuff sizes 3.5–11 cm**; large sizes bladder neck
- **Cuff slightly tighter than tape**; looser cuffs → better continence
- **3.5-cm cuff + RT → erosion**; prefer TC ± 51–60 PRB
- **PRB fill 23 mL**; contrast diluted with sterile water; **no contrast if iodine allergy**
- **Ectopic or inguinal PRB contraindicated**: mesh hernia repair, cystectomy, extensive abdominal surgery
- **Closed cuff → slight blanching** on urethroscopy
- **Tandem cuff**: ≥1.5–2 cm distal; Y-connector; +3 mL; off 6 weeks; not first-line
- **Single-cuff failure up to 11%**
- **TC**: **urethra <3.5 cm**; more retention; erections usually kept
- **Wilson transscrotal**: one scrotal incision; inferior continence
- **Bladder neck AUS**: **cuff ≥8 cm**, **PRB 71–80**; neuropathic (myelomeningocele); **contraindicated after RP**

MALE SLINGS

- **Transobturator:** Rehder and Gozzi 2007; after RP, TURP or laser enucleation
- **Trocar outside-in**, 1 fingerbreadth below adductor longus; **two pops**
- **Tension under cystoscopy to coaptation**; bulb moves 3–4 cm
- **ATOMS:** bulbospongiosus **not opened**; port 9 mL

PERSISTENT INCONTINENCE

- **Sudden** → deactivation or leak; **gradual** → atrophy
- **Reactivate:** hold button ≥30 s, squeeze pump edges
- **PRB >20 mL on imaging** → cuff problem; empty PRB → replace device
- **Sling loosened in first 6 weeks** → re-explore, replace
- **Failed sling** → AUS = gold standard; 2nd Advance ~56%, only if AUS refused

AUS COMPLICATIONS

- **Retention:** confirm deactivated → catheter 24–48 h → suprapubic catheter (image-guided); weeks → cuff undersized
- **Infection 1–3%** (10% high-risk); *Staphylococcus epidermidis*, *S. aureus*; late >4 months
- **Earliest sign:** scrotal pain; explant almost always
- **InhibiZone** coating: no proven AUS infection benefit
- **Salvage contraindicated:** sepsis, ketoacidosis, necrotising infection, erosion, immunosuppression, pus
- **Erosion 2–15%** (35% with multiple risk factors); stricture after 8–35%
- **Erosion <50% urethral loss** → catheter; ≥50% unresolved
- **Erosion** → urethral catheter ~3 weeks; new cuff away from old site
- **Re-erosion up to 35%**; nightly deactivation after reimplant
- **Atrophy:** downsize or relocate, TC, tandem
- **Device life 7–10 years**; <3 years → replace failed part (pump or cuff)

SPECIAL CIRCUMSTANCES

- **ROBUST III:** paclitaxel balloon **83.2% vs 21.7%** 1-year freedom from reintervention
- **Scope 22–24 Fr proximal to cuff** → uncouple cuff

- **VUAS <12 Fr:** dilate at the same time as AUS — no penalty
- **Neobladder SUI 33–95%**; AUS revision **60%**; PRB behind rectus
- **Dual AUS + penile prosthesis:** synchronous = metachronous; staged → **AUS first**
- **Dual implants revised more** than either alone
- **Mini-jupette:** 3-cm corporotomies + graft at penile prosthesis — climacturia and mild SUI

SLING COMPLICATIONS

- **Transobturator:** pain and wound commonest; **erosion 0–0.4%**; chronic pain 1.3%
- **ATOMS:** erosion and infection ~10%; **explant 10–16%**
- **ProACT:** **perforation ~5% commonest**; migration ~6.5%; revision 22%

SUMMARY

- **~90% improve in quality of life** after surgery, whatever the severity
- **>150,000 men** implanted

CLASSIC TRAPS

- **Trap:** transobturator sling compresses — **Actually:** repositions; needs a functioning sphincter
- **Trap:** sling after RT or severe UI — **Actually:** AUS first-line
- **Trap:** atrophy → raise PRB pressure — **Actually:** downsize or relocate cuff; higher PRB erodes
- **Trap:** urethral catheter for AUS retention — **Actually:** suprapubic after second failed trial
- **Trap:** bladder neck AUS after RP — **Actually:** contraindicated
- **Trap:** tandem cuff first-line — **Actually:** distal cuff erodes; wastes revision sites
- **Trap:** 3.5-cm cuff for irradiated atrophy — **Actually:** TC ± 51–60 PRB
- **Trap:** ProACT's commonest complication is erosion — **Actually:** perforation
- **Trap:** treat infected AUS with antibiotics — **Actually:** explant; salvage only if non-eroded
- **Trap:** dry after AUS = safe bladder — **Actually:** poor compliance may silently harm upper tracts